Client c

Item text

Frozen text encoder

x_i

Frozen semantic prior

① PGDR

Popularity-gated routing

Training item frequency

Source-initialized

Target-initialized

Head path

Tail path

Frequency-weighted mix

Mixed

Both paths train on target

② Private-profile FiLM

Client conditioning

Private profile p_c

Shared conditioner

Scale + shift

Head residual

Conditioned

Residual mix + normalize

③ AutoAnchor

Catalog-level anchoring

/

Users / items

Fixed catalog weight

Adapted feature

Frozen prior

Weighted interpolation

Normalize

Anchored item

Upload θ_c *only*

Broadcast θ

Server

Shared parameters

θ_h θ_l φ

FedAvg

Client models

Global θ

Equal-client average

Local BPR training

Local interactions

Sample a pair

Observed

Unobserved

User

BPR ranking objective

Observed

Unobserved

Prefer observed items

Local parameter updates

Shared θ_c

Private e_u

Private state

Kept on each client